Below is the finalized implementation document for the proposed solution.

Edit

AI-Powered Learning & Self-Test Interview Coach

Solution Architecture & Implementation Plan

Version: 1.0
Document Type: Technical Solution Architecture & Implementation Plan

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1. Executive Summary

The **AI-Powered Learning & Self-Test Interview Coach** is an Agentic AI platform that enables learners to prepare for technical interviews through personalized learning, adaptive assessments, and AI-driven mock interviews.

The solution combines three modern AI capabilities:

- **Retrieval-Augmented Generation (RAG)** for grounded knowledge retrieval.
- **Model Context Protocol (MCP)** for secure integration with enterprise tools and external systems.
- **Agentic AI** for autonomous planning, task orchestration, and multi-step execution.

To improve reliability, the solution also includes a **Question Bank Service** that allows trainers or administrators to upload interview questions using Excel or CSV files. This service acts as a fallback whenever RAG or MCP services are unavailable and also enables easy management of interview content without requiring AI expertise.

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2. Business Objectives

The solution should enable learners to:

- Learn technical concepts interactively.
- Practice coding and theoretical questions.
- Conduct AI-driven mock interviews.
- Receive personalized feedback.
- Track learning progress.
- Schedule future interview sessions.
- Prepare for company-specific interview patterns.

The platform should enable trainers and administrators to:

- Upload interview question banks.
- Manage interview content.
- Track learner performance.
- Create learning paths.
- Generate reports and analytics.

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3. Functional Requirements

Learner

- AI Chat Assistant
- Personalized Learning
- Coding Practice
- Mock Interviews
- Self Assessments
- Adaptive Question Difficulty
- Interview Feedback
- Progress Dashboard
- Interview History

Trainer

- Upload Excel Question Bank
- Upload CSV Question Bank
- Manage Questions
- Categorize Questions
- Version Questions
- View Student Reports

Administrator

- Manage Users
- Manage Question Banks
- Configure MCP Servers
- Manage Knowledge Base
- Monitor AI Performance
- Manage Security
- Audit Logs

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4. Solution Architecture

```
                       User
                        │
                        ▼
              Web / Mobile Application
                        │
                        ▼
                 API Gateway (FastAPI)
                        │
                        ▼
              Agentic AI Orchestrator
                        │
      ┌───────────────┬───────────────┬───────────────┬────────
      │               │               │               │
      ▼               ▼               ▼               ▼
Planning Agent    RAG Engine      MCP Layer      Question Bank S
      │               │               │               │
      │               │               │               │
      ▼               ▼               ▼               ▼
Learning Agent  Vector DB     GitHub MCP       Excel Upload
Quiz Agent      Embeddings    Calendar MCP     SQL Database
Interview Agent Documents     LMS MCP          Search Service
Evaluation      Knowledge     Email MCP
Feedback Agent  Repository    PostgreSQL MCP
                             Filesystem MCP
                        │
                        ▼
                 Large Language Model
```

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5. Recommended Technology Stack

Layer	Technology
Frontend	Angular 18 / React
Backend	FastAPI (Python)
Authentication	OAuth2 / Azure AD / JWT
AI Orchestration	LangGraph or OpenAI Agents SDK
LLM	GPT-5.5
RAG Framework	LlamaIndex
Embedding Model	OpenAI text-embedding-3-large
Vector Database	ChromaDB (Development), Qdrant/Pinecone (Production)
Relational Database	PostgreSQL
Cache	Redis
Object Storage	Azure Blob Storage / AWS S3
Excel Processing	Pandas + OpenPyXL
MCP SDK	Official MCP Python SDK
Queue	RabbitMQ
Monitoring	Langfuse + OpenTelemetry + Grafana
Deployment	Docker + Kubernetes
CI/CD	GitHub Actions / Azure DevOps

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6. Core Components and Responsibilities

6.1 Planning Agent

Responsibilities

The Planning Agent is the central orchestrator.

It is responsible for:

- Understanding user intent
- Creating execution plans
- Selecting specialized agents
- Deciding whether RAG is required
- Calling MCP tools
- Invoking Question Bank Service
- Managing fallback logic
- Combining results into a final response

The Planning Agent never directly answers questions; it delegates tasks.

6.2 Learning Agent

Responsibilities

- Explain concepts
- Generate examples
- Create study plans
- Recommend learning paths
- Provide coding explanations

6.3 Quiz Agent

Responsibilities

- Generate MCQs
- Generate Coding Challenges
- Generate Scenario-Based Questions
- Adapt Question Difficulty
- Evaluate Quiz Responses

6.4 Interview Agent

Responsibilities

- Conduct Mock Interviews
- Technical Interviews
- HR Interviews
- Behavioral Interviews
- Company-Specific Interviews
- Ask Follow-Up Questions

6.5 Evaluation Agent

Responsibilities

- Score Answers
- Evaluate Coding Solutions
- Assess Communication
- Identify Knowledge Gaps
- Generate Interview Scores

6.6 Feedback Agent

Responsibilities

- Interview Summary
- Strength Analysis
- Weakness Analysis
- Learning Recommendations
- Readiness Score
- Personalized Improvement Plan

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7. RAG Architecture

Purpose

Provide contextual knowledge retrieved from trusted learning resources.

Knowledge Sources

- Course Material
- PDF Documents
- Programming Guides
- Company Interview Experiences
- Coding Standards
- Internal Documentation
- Books
- Technical Blogs
- Sample Projects

RAG Workflow

1. User submits query.
2. Query embedding is generated.
3. Vector database performs semantic search.
4. Top matching documents are retrieved.
5. Relevant chunks are added to the LLM prompt.
6. LLM generates a grounded response.

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8. MCP Integration

Purpose

Enable secure interaction with enterprise systems.

MCP Servers

- Filesystem MCP
- PostgreSQL MCP
- GitHub MCP
- Calendar MCP
- Email MCP
- LMS MCP

Example Operations

- Retrieve learner history
- Retrieve coding assignments
- Schedule interview
- Send interview report
- Store interview score

- Read learning resources

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9. Question Bank Service (New Component)

Purpose

Provide structured interview content independent of RAG.

Supports:

- Excel Upload
- CSV Upload
- Manual Question Entry
- SQL Storage
- Question Search
- Version Control

Sample Excel Columns

Category	Topic	Difficulty	Question	Expected Answer	Keywords	Compa

Workflow

1. Trainer uploads Excel.
2. Validate schema.
3. Detect duplicates.
4. Import into PostgreSQL.
5. Optionally generate embeddings.
6. Make questions available to the Planning Agent.

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10. Hybrid Retrieval Strategy

The Planning Agent follows this retrieval priority:

Level 1

RAG

Used for:

- Learning content
- Documentation
- Explanations
- Concept retrieval

Level 2

MCP

Used for:

- Learner profile
- Calendar
- GitHub
- LMS
- Database
- Email

Level 3

Question Bank Service

Used when:

- RAG returns insufficient results.
- Interview questions are required.
- Structured assessments are preferred.
- Trainers have uploaded curated questions.

Level 4

LLM Knowledge

Used only when external sources are unavailable.

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11. Decision Flow

```
User Request
      |
      ▼
Planning Agent
      |
```

```
Need Learning?
      |
      ├── Yes → RAG
      |
      └── No
            |
Need Enterprise Data?
      |
      ├── Yes → MCP
      |
      └── No
Need Interview Questions?
      |
      ├── Yes → Question Bank
      |
      └── No
LLM Knowledge
```

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12. End-to-End Workflow

Example Request

"I want to prepare for a Python interview."

Execution:

1. Planning Agent understands intent.
2. Retrieve learner history via MCP.
3. Retrieve Asyncio learning material via RAG.
4. Learning Agent teaches the concept.
5. Quiz Agent generates practice questions.
6. Interview Agent starts a mock interview.
7. Evaluation Agent scores responses.
8. Feedback Agent generates recommendations.
9. Save progress through PostgreSQL MCP.
10. Schedule next session through Calendar MCP.
11. Send report using Email MCP.

If RAG is unavailable:

- Retrieve interview questions from the Question Bank Service.
- Continue interview workflow without interruption.

13. Development Roadmap

Phase 1 – Foundation

Deliverables

- Frontend
- Backend APIs
- Authentication
- PostgreSQL
- Docker
- CI/CD

Phase 2 – Knowledge Management

Deliverables

- Document ingestion
- Chunking
- Embeddings
- Vector database
- RAG APIs
- Question Bank Service
- Excel upload
- CSV upload
- SQL storage

Phase 3 – MCP Integration

Deliverables

- Filesystem MCP
- PostgreSQL MCP
- Calendar MCP
- GitHub MCP
- LMS MCP
- Email MCP

Phase 4 – Agentic AI

Deliverables

- Planning Agent
- Learning Agent
- Quiz Agent
- Interview Agent
- Evaluation Agent
- Feedback Agent
- Multi-agent orchestration
- Fallback logic

Phase 5 – User Experience

Deliverables

- AI Chat
- Learning Dashboard
- Coding Playground
- Mock Interview
- Reports
- Trainer Portal
- Question Bank Management

Phase 6 – Production Readiness

Deliverables

- Monitoring
- Security
- Performance Testing
- Prompt Evaluation
- Logging
- Backup & Recovery
- High Availability

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14. Responsibility Matrix

Component	Responsibility
Frontend	User interaction and dashboards
FastAPI Backend	APIs, authentication, sessions
Planning Agent	Workflow orchestration
Learning Agent	Concept explanation
Quiz Agent	Question generation
Interview Agent	Mock interview execution
Evaluation Agent	Scoring and assessment
Feedback Agent	Reports and recommendations
RAG Engine	Knowledge retrieval
Embedding Service	Vector generation
Vector Database	Semantic search
MCP Layer	Enterprise integrations
Question Bank Service	Structured interview repository
PostgreSQL	Learner and question data
Redis	Caching
Monitoring	Observability and tracing

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15. Non-Functional Requirements

- Secure authentication and authorization
- High availability for AI services
- Horizontal scalability
- Low-latency semantic search
- Audit logging
- Role-based access control

- Encryption in transit and at rest
- Configurable prompts and AI models
- Monitoring and alerting
- Disaster recovery and backup

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16. Benefits

- Personalized AI-powered learning
- Adaptive mock interviews
- Reliable enterprise integrations via MCP
- Context-aware responses using RAG
- Excel-based question management for non-technical users
- Resilient fallback mechanism through the Question Bank Service
- Modular, scalable, and maintainable architecture
- Enterprise-ready deployment model

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17. Implementation Summary

The implementation follows a modular Agentic AI architecture where the **Planning Agent** orchestrates all workflows. **RAG** provides contextual knowledge from curated learning content, **MCP** enables secure integration with enterprise systems such as LMS, calendars, GitHub, and databases, while the **Question Bank Service** ensures uninterrupted interview functionality through curated Excel/CSV-based question repositories.

This hybrid approach provides:

- Intelligent planning through specialized AI agents.
- Grounded and explainable responses using RAG.
- Secure external system integration using MCP.
- Operational resilience through a structured Question Bank fallback.
- A scalable foundation for future enhancements such as voice interviews, coding execution sandboxes, multilingual support, and advanced analytics.

The resulting solution is suitable for enterprise training platforms, university learning systems, corporate upskilling programs, and AI-assisted interview preparation applications.

This document is suitable as a **Solution Architecture / Technical Design Document (TDD)** for implementation. It can be further extended with UML diagrams (Use Case, Sequence, Component, Deployment), database ER diagrams, API specifications (OpenAPI), LangGraph workflow diagrams, and sprint-wise implementation tasks for an engineering team.